Coverage-Oriented Planner-Guided Memory Residual Reinforcement Learning for Robotic Thermal-Spot Coverage

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***Abstract*—Localized steam breakthroughs in a granular thermal-processing vessel appear online and remain active until they are covered. We study this persistent service problem in a MuJoCo digital twin of an ABB manipulator, where an external thermal localization module supplies clustered target positions and the robot moves a coverage center over the active spots. A Horizon-2 receding-horizon planner provides a structured action prior, while an LSTM-PPO policy learns a phase-scaled, shielded residual from the active set, spawn history, thermal context, and route summaries. We use three training seeds, three held-out environment seeds, and five episodes per crossed cell. Relative to Horizon-2, coverage is statistically tied in Low, Realistic, and Hard; in Extreme it increases from 0.816 to 0.855, with a hierarchical 95% interval for the difference of $[0.003, 0.074]$, but the four-stage Holm-adjusted test is not significant. A matched absolute-action ablation that retains the encoder, warm start, prediction loss, and service reward reduces coverage by 0.079–0.168 across the four stages. The Extreme gain is accompanied by higher latency and backlog. The result is therefore a coverage-oriented planner-learning operating point, not a universally faster controller.**

***Index Terms*—robotic coverage, residual reinforcement learning, persistent service, coverage path planning, LSTM-PPO, MuJoCo**

I. Introduction

This work is motivated by robot-assisted loading and steaming of a granular bed. Localized steam breakthroughs are detected by an external thermal sensing system, and an ABB manipulator moves a deposition or coverage center over the vessel to serve each active location. The simulator supplies target coordinates, so the paper evaluates online control and service routing rather than camera perception. Targets arrive in spatially clustered burst–lull cycles, remain active until covered, and can accumulate while the arm is in transit. This makes the task a persistent service problem: geometric travel, waiting time, target age, and congestion must be considered together.

The problem is related to coverage path planning [1], coverage control [2], statistical coverage [3], and persistent monitoring with stochastic arrivals [4]. It also has a direct connection to stochastic dynamic vehicle routing, where online arrivals and waiting time determine service quality [5], [6]. Unlike static map coverage, an active spot leaves the queue only after physical coverage. Reinforcement learning has been applied to adaptive coverage planning [7], [8], but a visible-target planner is already a strong prior in this task. Residual reinforcement learning [9] and bounded residual control [10], [11] provide a conservative way to retain that prior. The set encoder follows the permutation-invariant motivation of Set Transformer [12].

The paper is primarily aligned with ICCC Track 6 (Robotics), with secondary links to Track 2 (AI for Engineering) and Track 4 (Machine Learning). We ask: *RQ1*: does a bounded residual action interface improve a fixed planner relative to a matched absolute-action PPO controller? *RQ2*: how does the method compare with alternative planner and heuristic priors? *RQ3*: what coverage–service trade-off is created in latency, strict SLA, backlog, and smoothness?

Our claims are intentionally narrow. Learning does not replace planning; it learns a correction around a competent online route prior. The contributions are (i) a persistent thermal-spot service formulation with explicit metric denominators, (ii) an LSTM-PPO residual interface with attention, phase-scaled authority, and a progress shield, and (iii) a crossed held-out evaluation whose uncertainty and component accounting separate the residual interface from auxiliary modules.

II. Task and Method

A. Persistent service and metrics

Let $\mathcal{S}_t = \{p_i(t)\}$ be the active spot set, where $p_i = (x_i, y_i, a_i)$ contains location and active age, and let $c_t = (x_t, y_t)$ be the coverage center. Spot i is served when

$$\|c_t - (x_i, y_i)\|_2 \le r_{\text{cover}}. \tag{1}$$

The spot is then removed. Timeout removal is disabled in the reported protocol; an uncleared spot remains in the denomina-

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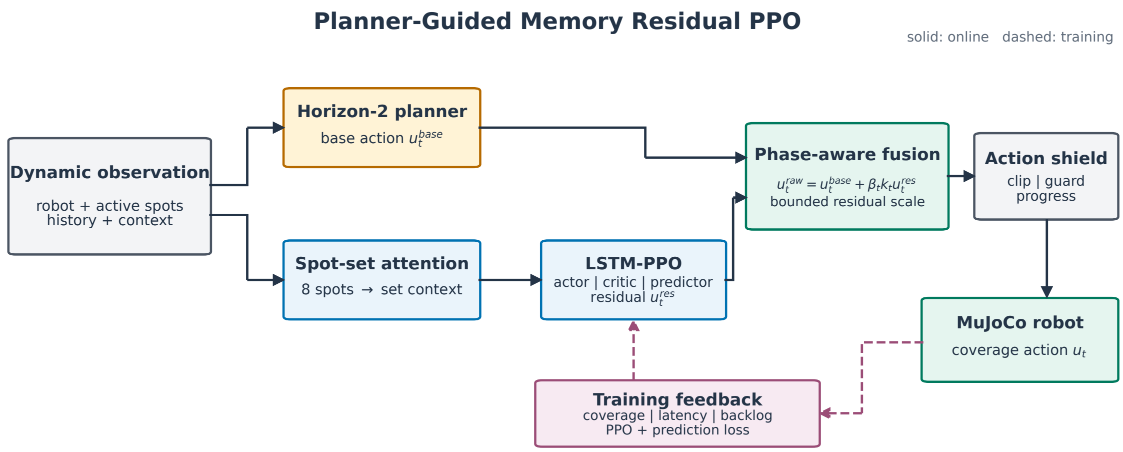


Fig. 1. Planner-guided residual control. The online planner remains in the action loop; LSTM-PPO supplies a bounded correction and the shield checks local progress before the action is sent to the MuJoCo robot.

tor. The action $u_t = [\Delta x_t, \Delta y_t, s_t] \in [-1, 1]^3$ is issued every 0.05 s. Coverage is

$$C = \frac{N_{\text{covered}}}{N_{\text{spawned}}}. \tag{2}$$

For a covered spot, $T_i = t_i^{\text{cover}} - t_i^{\text{spawn}}$ and $\bar{T} = N_{\text{covered}}^{-1} \sum_i T_i$. Thus mean and p90 latency are conditional on covered spots. Strict SLA exposes the uncleared tail:

$$C_{\text{SLA}} = \frac{|\{i : T_i \leq T_{\text{SLA}}\}|}{N_{\text{spawned}}}, \qquad T_{\text{SLA}} = 4.0 \text{ s}. \tag{3}$$

Backlog is the time-average active count and smoothness is $\Delta_u = T^{-1} \sum_t \|u_t - u_{t-1}\|_2$. Coverage is the primary objective; the other quantities describe its service cost.

B. Planner prior and recurrent residual

The interpretable risk score used by the comparator and as a diagnostic is

$$q_i = 0.40\phi_a + 0.30\phi_d + 0.10\phi_r + 0.20\phi_h, \tag{4}$$

where ϕ_a is normalized age, ϕ_d normalized closeness, ϕ_r normalized reachability, and ϕ_h local thermal score, all clipped to their stated ranges. The risk-aware baseline greedily selects the maximum-q_i target; ties use target index.

Horizon-L enumerates ordered routes of length $\min(L, N_t)$. For route $\rho = (i_1, \ldots, i_L)$, the planner simulates travel in steps and uses

$$\begin{aligned} g_j &= q_{i_j} + 0.45 z_j - 0.018 d_j - 0.050 (z_j - 1)_+, \\ J(\rho) &= \sum_{j=1}^{L} 0.82^{j-1} g_j - 0.08 \bar{z}_{\text{left}}, \end{aligned} \tag{5}$$

where d_j is travel steps and z_j is arrival age normalized by the stage age scale. H2 uses $L = 2$ and executes the first action of the highest scoring route; H3 uses the identical objective with $L = 3$. Nearest, oldest, distance–age, dynamic-weighted, ACO-TSP, and planner-ensemble baselines are defined by their names plus the same bounded workspace and service rule.

The observation has $61 + 8 \times 8 = 125$ features. Eight spot slots are embedded with a shared width-128 map and four-head masked attention; the global branch and set vector feed a one-layer width-256 LSTM. Actor, critic, and a two-dimensional spawn-prediction head share this recurrent representation. The hidden state is carried across consecutive rollout chunks and coverage events. With planner action u_t^{base} and residual $u_t^{\text{res}} = \pi_\theta(o_t, h_t)$, the candidate is

$$\begin{aligned} \tilde{u}_t &= \text{clip}(u_t^{\text{base}} + \beta_t k_t u_t^{\text{res}}, -1, 1), \\ \beta_t &= 0.03 + 0.17\, \text{clip}(n_t / 700000, 0, 1). \end{aligned} \tag{6}$$

The phase multipliers (k_t) for lull, sparse, charging, mid, dense, and burst are $(1.35, 1.25, 1.00, 1.00, 0.40, 0.25)$. The post-policy shield rejects poor alignment, requires at least 0.35 of base one-step progress, and uses a 0.70-base/0.30-candidate guarded blend with a 0.50 progress ratio. It is a local progress filter, not a formal temporal-logic safety guarantee [13].

The scalarized reward contains time, active-count, mean-age, potential, best-progress, coverage, quick-cover, covered-latency, oldest-active, backlog, SLA, action-change, action norm, and terminal success/miss terms. The load-bearing coefficients are $w_{\text{time}} = (.030, .035, .045, .055)$, $w_{\text{active}} = .024$, $w_{\text{age}} = .112$, $w_{\text{cover}} = .70$, $w_{\text{quick}} = 3.0$, $w_{\text{precision}} = 7.8$, $w_{\text{potential}} = .65$, $w_{\text{progress}} = .3375$, $w_{\text{latency}} = 30$, $w_{\text{oldest}} = .45$, $w_{\text{backlog}} = .20$, $w_{\text{SLA}} = (20, -30)$, $w_{\Delta u} = .025$, $w_{\|u\|} = .004$, and terminal $(+30, -28)$. The reward is multiplied by 0.02 and clipped to $[-4, 4]$; hence SLA and backlog are preferences, not hard CMDP constraints. Constrained Policy Optimization [14] is a relevant alternative, but is not used.

III. Experiments

A. Protocol and statistical unit

The simulator is a MuJoCo digital twin [15] containing an ABB arm, a circular vessel, and an externally localized target

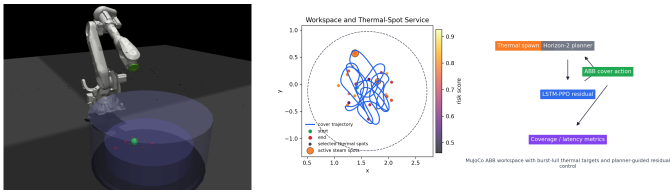


Fig. 2. MuJoCo ABB workspace (left) and online thermal-spot service task (right). The controller moves the green coverage center over clustered spots inside the circular vessel.

TABLE I
STAGE AND TRAINING PARAMETERS

Stage	$N_{\max}$	r_{cover}	A_{stage}	p_{spawn}	cooldown
Low	3	.17	260	.006	80
Realistic	4	.14	220	.010	65
Hard	6	.12	200	.018	48
Extreme	8	.10	180	.026	36
Obs. / widths	125; spot 128, global/LSTM 256; 4-head attention				
Sequence / update	$4 \times 800 = 3200$; PPO update every 4 chunks				

field. Three drifting hotspots generate burst–lull arrivals (70-step lull, 110-step charging, then 3–5 spots one at a time every four steps); background weight is 0.055 and hotspot strength is 3.8. Table I gives the four load stages. Each learned method uses training seeds 0–2 and disjoint evaluation seeds 100–102; each crossed cell has five independent 3,200-step episodes. The final fixed-curriculum checkpoint is evaluated and test seeds are never used for selection. Training uses 921,600 environment steps per seed, PPO $(\gamma, \lambda) = (.985, .95)$, learning rate 10^{-5}, one epoch, clip 0.05, gradient norm 0.35, H2 behavior-cloning warm start for 200 episodes, and prediction coefficient 0.05 with a 150-step horizon.

We use 20,000 crossed bootstrap draws: training labels, evaluation labels, and episodes are resampled at their respective levels, while paired methods reuse the same held-out scenarios. Tables report hierarchical 95% intervals; Holm correction is applied across the four stages for each predeclared comparison family. This keeps training randomness and environment randomness from being treated as 45 independent flat observations.

B. Coverage and baselines

Table II is the primary comparison. Ours is tied with H2 in the first three stages. Extreme has the clearest numerical separation, but its uncorrected interval does not survive four-stage Holm correction. The full baseline landscape in Table III prevents an overstrong claim: risk-aware is slightly higher than ours in Extreme, while H3 and the planner ensemble are competitive in selected stages.

C. Action-interface and component evidence

The matched no-residual policy retains the same 125-dimensional observation, LSTM, H2 behavior-cloning warm start, prediction loss, service reward, and training budget. It emits an absolute action, with no planner base or shield. This is the causal action-interface comparison; the historical Vanilla LSTM-PPO run simultaneously removed residual composition, BC, and prediction, so it is not used to claim that LSTM itself fails. Table IV(a) reports the matched effect, and panel (b) gives exact component accounting.

TABLE II
COVERAGE VERSUS HORIZON-2 (HIERARCHICAL 95% CIS)

Stage	Ours	H2	Δ	Δ CI	p_{Holm}
Low	.917 [.906,.929]	.919 [.906,.932]	-.002	[-.018,.016]	1.000
Real.	.901 [.884,.922]	.896 [.883,.910]	+.005	[-.018,.034]	1.000
Hard	.870 [.844,.894]	.870 [.847,.894]	$<$ .001	[-.040,.033]	1.000
Extreme	.855 [.828,.881]	.816 [.782,.849]	+.038	[.003,.074]	.135

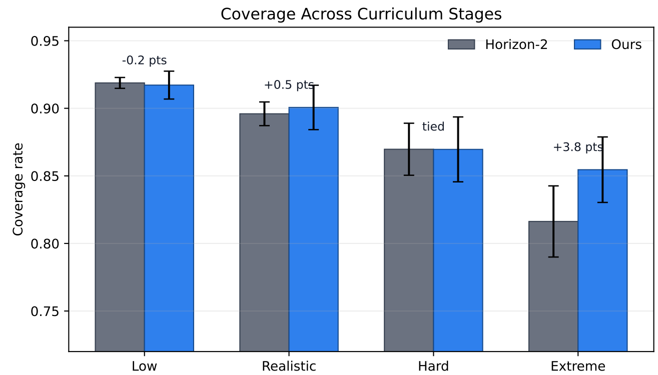


Fig. 3. Stage-wise coverage with hierarchical 95% intervals. The separation is concentrated in Extreme and is not corrected-significant.

D. Service diagnostics and robustness

Table IV(c) reports the quantities that coverage alone hides. Ours is close to H2 in Low–Hard. In Extreme, the 0.038 coverage increase comes with 2.50 s higher mean latency, 6.09 s higher p90, 0.26 more active spots, and lower strict SLA. The reward sweep therefore changes the operating point rather than removing the coverage–service trade-off. On an unseen higher-density condition, ours obtains .868/.850 in Hard/Extreme and H2 obtains .888/.825; the intervals overlap, so this is robustness evidence, not dominance. H3 is included as a planner-capacity comparator; a separately retrained H3-residual controller is outside the present five-page revision and is not claimed.

E. Qualitative rollout

Figure 4 compares one held-out Realistic episode under the same target-generation protocol. Ours/H2 have path lengths 15.37/16.40 and 25/40 selected-target switches. This example is interpretive only; aggregate claims come from the crossed-seed tables.

IV. DISCUSSION AND CONCLUSION

The strongest supported conclusion is about the action interface, not universal planner replacement. Across all stages, removing the planner-residual composition lowers coverage under the matched protocol, with corrected intervals excluding

TABLE III
MEAN COVERAGE ACROSS THE DETERMINISTIC AND LEARNED BASELINE SET

Method	Low	Real.	Hard	Extreme	Definition
Ours	.917	.901	.870	.855	H2 + bounded LSTM residual
Horizon-2	.919	.896	.870	.816	route enumeration, $L = 2$
Horizon-3	.917	.927	.860	.841	route enumeration, $L = 3$
Nearest	.922	.887	.860	.789	minimum distance
Oldest	.914	.890	.883	.807	maximum age
Distance–age	.921	.896	.851	.801	age / distance
Risk-aware	.924	.901	.836	.863	maximum q_i
Dynamic weighted	.927	.886	.865	.799	phase-weighted heuristic
ACO-TSP / ensemble	.925/.935	.893/.899	.864/.870	.754/.814	route / planner mix

TABLE IV
MECHANISM AND SERVICE EVIDENCE: (A) MATCHED RESIDUAL-INTERFACE EFFECT, (B) COMPONENT ACCOUNTING, AND (C) COVERAGE–SERVICE DIAGNOSTICS.

(a) Matched residual-interface effect

Stage	Full	No residual	Δ	Δ CI	p_{Holm}
Low	.917	.838 [.712,.915]	+.079	[.004,.202]	.029
Real.	.901	.776 [.667,.865]	+.124	[.036,.223]	.003
Hard	.870	.726 [.590,.837]	+.144	[.037,.268]	.011
Extreme	.855	.687 [.585,.793]	+.168	[.050,.285]	.003

(b) Exact component accounting

Variant	Action	Attn.	Carry	Pred.	Svc.	Shield
Full	H2+res.	Y	Y	Y	Y	Y
No residual	absolute	Y	Y	Y	Y	N
No attention	H2+res.	N	Y	Y	Y	Y
No carry	H2+res.	Y	N	Y	Y	Y
No prediction	H2+res.	Y	Y	N	Y	Y
No service	H2+res.	Y	Y	Y	N	Y

(c) Coverage and service diagnostics versus H2

Stage	Method	Cov.	Mean lat. (s)	p90 (s)	Strict SLA	Backlog	Δ_u
Low	Ours	.917	12.54	24.15	.181	2.30	.0254
	H2	.919	12.16	23.72	.191	2.28	.0255
Real.	Ours	.901	16.04	29.70	.147	2.84	.0261
	H2	.896	16.52	29.45	.131	2.82	.0258
Hard	Ours	.870	19.59	38.10	.127	3.49	.0277
	H2	.870	19.23	35.44	.121	3.43	.0274
Extreme	Ours	.855	23.88	46.01	.103	3.81	.0289
	H2	.816	21.38	39.92	.128	3.55	.0277

Mean and p90 latency condition on covered spots; strict SLA uses all spawned spots. Backlog is mean active count and Δ_u is mean per-step action change.

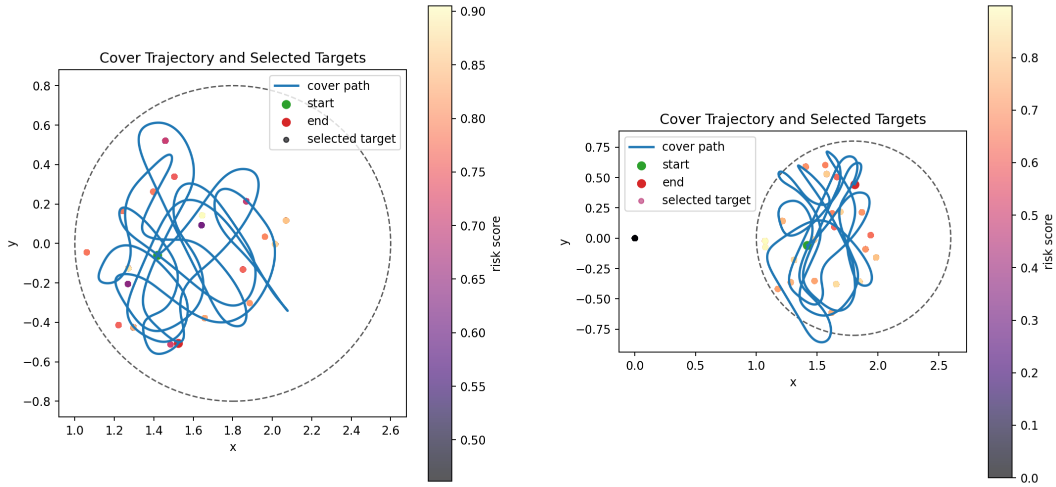
zero. Against H2 itself, the method is tied in three stages and has a positive but uncorrected Extreme difference. The full baseline table further shows that the risk-aware heuristic can be better in Extreme coverage, although with longer service time. These results support a configurable planner-learning architecture rather than a claim of universal superiority.

The attribution is narrower than a comparison with the historical direct LSTM-PPO run would suggest. That run changed the action interface, removed the H2 behavior-cloning warm start, and disabled the prediction objective at the same time. Its low coverage therefore cannot identify residualization as the cause. The matched no-residual controller is the appropriate mechanism test because it retains the observation, recurrent encoder, warm start, prediction loss, reward, curriculum, and optimization budget. The auxiliary ablations have mixed stage-wise effects; attention, recurrent carry, prediction, and service shaping are supporting design choices, not four independently proven contributions.

Planner-horizon evidence requires the same care. H3 is a strong deterministic comparator and shows that longer lookahead is not uniformly better: it improves Realistic and Extreme relative to H2, but not Low or Hard. However, the learned residual was trained around H2 actions. The H3 row therefore tests planner capacity, not transfer of the learned correction to another base. A matched H3-residual training run is required before claiming horizon-independent residual gains; the present paper does not make that claim.

The service metrics also clarify why coverage and response speed cannot be reported as one number. An active spot that is eventually covered contributes to coverage and covered-only latency; an uncleared spot contributes to the coverage denominator and strict-SLA failure but has no finite cover latency. The 0.05 s interval converts task age to seconds and is not a measurement of network inference time. Future work should use an explicit CMDP or constrained assignment when deadlines are hard requirements, and should train a matched H3-residual variant and validate the controller with thermal-camera noise and real robot timing.

In summary, this paper presents a reproducible ABB/MuJoCo persistent-service benchmark and a planner-guided LSTM-PPO residual controller. The residual interface is the most defensible mechanism-level result; the coverage gain in Extreme is useful but statistically bounded and paid for with latency and backlog. The revised evidence and definitions are intended to make that trade-off explicit and reproducible.



(a) Planner-guided residual PPO

(b) Horizon-2

Fig. 4. Representative held-out Realistic trajectories; aggregate claims use all crossed training/evaluation seed cells.

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